

E086

Computer Systems for AI

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A research group at a university with a famous football team is training a convolutional network whose time is spent almost entirely in large matrix multiplications. Their budget was approved in June. Which hardware choice best speeds up training?

- A. More system RAM on the same CPU.
- B. Faster SSD storage.
- C. A CPU with a higher clock speed.
- D. GPUs or other accelerators built for parallel matrix operations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A startup with offices in Austin wants to serve a 7-billion-parameter model in FP16. Its founders previously worked at a chip company. About how much memory do the weights alone need?

- A. About 7 GB
- B. About 28 GB
- C. About 14 GB
- D. About 112 GB

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A team can spend at most 20 GB of GPU memory on weights and will store them in INT8. What is the largest model, in parameters, that fits?

- A. About 10 billion
- B. About 160 billion
- C. About 5 billion
- D. About 20 billion

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A chatbot runs a 14 GB model at batch size 1 on a GPU with 2 TB/s memory bandwidth and 300 TFLOPS of peak compute. The GPU vendor recently changed its logo. About what is the upper bound on tokens generated per second?

- A. About 2,000 tokens/s
- B. About 21,000 tokens/s
- C. About 143 tokens/s
- D. About 0.007 tokens/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A 20 GB model must generate 50 tokens per second at batch size 1. About how much memory bandwidth is needed at minimum?

- A. About 2.5 GB/s
- B. About 20 GB/s
- C. About 1,000 GB/s
- D. About 70 GB/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A training job at a biotech firm with a new rooftop garden is 80% parallelizable. The team moves from 1 GPU to 8 GPUs purchased at a year-end discount. What is the best-case overall speedup?

- A. 5×
- B. About 3.3×
- C. 6.4×
- D. 8×

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A job is 90% parallelizable. What is the maximum speedup possible no matter how many processors are added?

- A.** About 1.1×
- B.** 10×
- C.** Unlimited
- D.** 9×

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A profiling run on a new 8-GPU server, set up by a contractor from a nearby city, shows GPU utilization at 35% while every CPU core is at 100% decoding JPEG images in the data loader. The server room is kept at 18 °C. What should the team do first?

- A.** Speed up the input pipeline: more loader workers, prefetching, or pre-decoded data.
- B.** Buy faster GPUs.
- C.** Low GPU utilization always means the model is too small.
- D.** Make the model larger to use the GPUs fully.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A food-packaging plant, owned by a family business since 1962, needs a camera to reject bad seals within 20 ms. The plant's internet drops several times a day and the device has a 10-watt power budget. Which deployment fits best?

- A.** Run the largest available model on the device.
- B.** On-device inference with a small, quantized model.
- C.** Send each image to a large cloud model.
- D.** Stream video to the cloud and cache results locally.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

The recommendation model behind an online bookstore with a cat mascot times out during a holiday traffic spike. The model was trained on five years of orders. What is the best safe-degradation behavior?

- A. Show an error page until the model returns.
- B. Retry the model call until it answers.
- C. Nothing; a well-tested model never fails.
- D. Time out quickly and show a simple fallback such as best-sellers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A triage-support model receives an input very unlike anything in its training data. What is the safest behavior?

- A. Abstain and route the case to a person for review.
- B. Crash the application.
- C. Return its best guess with high confidence.
- D. Silently drop the request.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A model service that runs fine on a developer's laptop, which has a large sticker collection, fails in production with a CUDA library version error. The team has shipped 30 releases this year. What prevents this class of problem?

- A. Build a container image with pinned versions and deploy that same image everywhere.
- B. Move to a larger production server.
- C. Always install the latest versions on each server.
- D. Write the correct versions on a wiki page.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A research group at a university ~~with a famous football team~~ is training a convolutional network whose time is spent almost entirely in large matrix multiplications. ~~Their budget was approved in June.~~ Which hardware choice best speeds up training?

Target word: hardware for parallel matrix math **Decides:** Work is dominated by matrix multiplication.

A	Adjacent category	Capacity does not add parallel compute.
B	Confusion set	Storage helps loading data, not compute-bound math.
C	Absolute language	Clock speed helps serial work; it cannot match massive parallelism.
D	Correct	Thousands of parallel units match matrix-heavy work.

2. Answer C Concept 2

A startup ~~with offices in Austin~~ wants to serve a 7-billion-parameter model in FP16. ~~Its founders previously worked at a chip company.~~ About how much memory do the weights alone need?

Target word: weight memory **Decides:** $7B \times 2$ bytes.

A	Dropped element	Uses 1 byte per parameter (INT8).
B	Adjacent category	Uses 4 bytes per parameter (FP32).
C	Correct	$7 \times 10^9 \times 2$ bytes = 14×10^9 bytes.
D	Confusion set	Treats 16 bits as 16 bytes.

3. Answer D Concept 2 · transfer

A team can spend at most 20 GB of GPU memory on weights and will store them in INT8. What is the largest model, in parameters, that fits?

Target word: solve for parameter count **Decides:** INT8 = 1 byte per parameter.

A	Adjacent category	Uses 2 bytes per parameter (FP16).
B	Inverted operation	Multiplies by 8 bits instead of treating one byte as one parameter.
C	Confusion set	Uses 4 bytes per parameter (FP32).
D	Correct	20×10^9 bytes $\div$ 1 byte each.

4. Answer C Concept 3

A chatbot runs a 14 GB model at batch size 1 on a GPU with 2 TB/s memory bandwidth ~~and 300 TFLOPS of peak compute.~~ ~~The GPU vendor recently changed its logo.~~ About what is the upper bound on tokens generated per second?

Target word: decode speed bound **Decides:** Each token reads all 14 GB of weights; bandwidth is the limit.

A	Wrong stem number	Reports the bandwidth figure in GB/s as a token rate.
B	Camouflage	Uses peak FLOPS; batch-1 decoding is bandwidth-bound.
C	Correct	$2,000 \text{ GB/s} \div 14 \text{ GB} \approx 143$.
D	Inverted operation	Divides model size by bandwidth.

5. Answer C Concept 3 · transfer

A 20 GB model must generate 50 tokens per second at batch size 1. About how much memory bandwidth is needed at minimum?

Target word: solve for bandwidth **Decides:** Bandwidth $\geq$ tokens/s $\times$ model bytes.

A	Inverted operation	Divides tokens by size instead of multiplying.
B	Wrong stem number	Repeats the model size.
C	Correct	$50 \times 20 \text{ GB} = 1,000 \text{ GB/s}$.
D	Adjacent category	Adds the two numbers.

6. Answer B Concept 4

A training job at a biotech firm ~~with a new rooftop garden~~ is 80% parallelizable. The team moves from 1 GPU to 8 GPUs ~~purchased at a year-end discount~~. What is the best-case overall speedup?

Target word: Amdahl speedup **Decides:** $1 / (0.2 + 0.8/8)$.

A	Confusion set	That is the limit with unlimited GPUs, not with 8.
B	Correct	$1 / (0.2 + 0.1) = 3.33$.
C	Adjacent category	0.8×8 is not how parallel and serial time combine.
D	Dropped element	Ignores the 20% serial part.

7. Answer B Concept 4 · transfer

A job is 90% parallelizable. What is the maximum speedup possible no matter how many processors are added?

Target word: solve for the speedup ceiling **Decides:** Limit = $1 / (1 - p)$.

A	Inverted operation	$1 / 0.9$ uses the parallel fraction.
B	Correct	$1 / 0.1 = 10$.
C	Absolute language	The serial 10% never shrinks.
D	Wrong stem number	Reads 90% as nine-fold.

8. Answer A Concept 5

A profiling run on a new 8-GPU server, ~~set up by a contractor from a nearby city~~, shows GPU utilization at 35% while every CPU core is at 100% decoding JPEG images in the data loader. ~~The server room is kept at 18 °C.~~ What should the team do first?

Target word: fix the real bottleneck **Decides:** GPUs are idle waiting on CPU-bound data loading.

A	Correct	The data pipeline is the slowest stage.
B	Inverted operation	The GPUs are already underused.
C	Absolute language	Here the profile clearly shows a CPU bottleneck.
D	Confusion set	Adds work without fixing the starved pipeline.

9. Answer B Concept 6

A food-packaging plant, ~~owned by a family business since 1962,~~ needs a camera to reject bad seals within 20 ms. The plant's internet drops several times a day and the device has a 10-watt power budget. Which deployment fits best?

Target word: deployment choice **Decides:** Tight latency, unreliable network, and low power.

A	Adjacent category	Right location, but breaks the 10-watt budget.
B	Correct	Meets latency and power limits without depending on the network.
C	Dropped element	Ignores network drops and the 20 ms deadline.
D	Confusion set	Each new seal still needs a cloud round trip.

10. Answer D Concept 7

The recommendation model behind an online bookstore ~~with a cat mascot~~ times out during a holiday traffic spike. ~~The model was trained on five years of orders.~~ What is the best safe-degradation behavior?

Target word: safe fallback **Decides:** A non-critical dependency failed under load.

A	Dropped element	Lets a non-critical feature take down the whole page.
B	Inverted operation	Adds load to an overloaded service.
C	Absolute language	Dependencies fail under load regardless of testing.
D	Correct	The page stays useful while the model recovers.

11. Answer A Concept 7 · transfer

A triage-support model receives an input very unlike anything in its training data. What is the safest behavior?

Target word: reverse: out-of-scope input to behavior **Decides:** Out-of-distribution input makes the prediction unreliable.

A	Correct	Escalates when the model cannot be trusted.
B	Adjacent category	Fails, but not safely or gracefully.
C	Inverted operation	Overstates reliability exactly when it is lowest.
D	Dropped element	No one learns the case needs attention.

12. Answer A Concept 8

A model service that runs fine on a developer's laptop, ~~which has a large sticker collection,~~ fails in production with a CUDA library version error. ~~The team has shipped 30 releases this year.~~ What prevents this class of problem?

Target word: repeatable environments **Decides:** Environments differ in library versions.

A	Correct	Every environment runs identical dependencies.
B	Adjacent category	Capacity does not fix a version mismatch.
C	Inverted operation	"Latest" changes over time, creating more drift.
D	Dropped element	Nothing enforces the documented versions.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Matching hardware to AI work	GPUs and TPUs excel at massively parallel matrix math; CPUs handle branching, orchestration, and serial work.	Choosing hardware by clock speed or core count alone.	A workload dominated by large matrix multiplications.
2	Model memory footprint	Weight memory $\approx$ parameters $\times$ bytes per parameter (FP32 = 4, FP16/BF16 = 2, INT8 = 1).	Mixing bits with bytes, or using the wrong precision.	A parameter count and a numeric precision.
3	Memory-bandwidth-bound inference	Single-stream token generation reads all weights per token, so $\text{tokens/s} \leq \text{memory bandwidth} \div \text{model bytes}$.	Estimating decode speed from peak FLOPS.	Batch size 1, a model size in GB, and a bandwidth in GB/s or TB/s.
4	Amdahl's law	$\text{Speedup} = 1 / ((1 - p) + p/n)$; the serial part caps speedup at $1 / (1 - p)$.	Expecting n processors to give $n \times$ speedup.	A parallel fraction and a processor count.
5	Profiling and bottleneck diagnosis	Measure first; the slowest stage sets the pace, and low accelerator utilization with busy CPUs points to the input pipeline.	Buying a faster GPU when the GPU is idle waiting on data.	Utilization numbers for GPU, CPU, disk, or network.
6	Cloud vs. edge deployment constraints	Tight latency, weak connectivity, privacy, and power limits push inference to the edge, usually with a smaller or quantized model.	Defaulting to the biggest cloud model regardless of the constraints.	Millisecond deadlines, watt budgets, or unreliable networks.
7	Failure modes and safe degradation	Plan what happens when a model or dependency fails: timeouts, fallbacks, and abstaining or escalating to people when inputs are out of scope.	Assuming a tested model will not fail, or retrying forever.	A model service outage or inputs unlike the training data.
8	Standardized runtime environments	Container images with pinned versions and configuration as code make deployments repeatable across machines.	Relying on documentation or "latest" versions to keep environments the same.	"Works on my machine" failures after deployment.

Course scope (public description): This course examines computer systems and architectures through the lens of AI engineering, emphasizing how hardware and software decisions shape the feasibility, performance, cost, and reliability of modern AI workloads. Learners analyze the roles of CPUs, GPUs, TPUs, memory hierarchies, storage, interconnects, and distributed systems in enabling efficient training and inference. The course emphasizes architectural tradeoffs, profiling and bottleneck diagnosis, and resource management strategies that influence latency, throughput, scalability, and operational constraints. Through applied analysis and system design activities, learners translate functional and non-functional requirements into concrete compute, memory, bandwidth, and power constraints across cloud, embedded, and edge environments. Learners evaluate failure modes and safe degradation behaviors and apply infrastructure foundations, including standardized runtime environments and repeatable configuration concepts, to support defensible deployment decisions. By the end of the course, learners can design and justify system architectures aligned to real-world AI production constraints and organizational objectives. pipelines, apply word embeddings, and use sequence-based and transformer-based models to complete tasks such as classification, sentiment analysis, and question answering. The course emphasizes rigorous evaluation beyond accuracy, using task-appropriate metrics, structured error analysis, and

stress testing under realistic input variation, to identify failure modes and reliability risks. Learners also assess bias, safety, and responsible-use concerns (including generative AI risks when applicable) and produce governance artifacts that document intended use, limitations, evaluation results, and monitoring expectations. Through applied engineering work, learners demonstrate judgment in selecting methods and designing solutions that meet technical, ethical, and operational constraints.